

2009-2010 秋季学期概率论与数理统计期末考试参考答案和评分细则

一、选择题 (10 分, 每空 2 分, 将选项对应的大写英文字母直接写在横线上)

题号	1	2	3	4	5
A 卷	A	A	B	B	C
B 卷	B	C	A	A	B

二、填空题 (15 分, 每空 3 分, 将计算结果直接写在横线上)

题号	6	7	8	9	10
A 卷	0.3	0.25	p	$2(n-1)/n$	2
B 卷	p	$2(n-1)/n$	2	0.3	0.25

注: A9B7 也写成以下形式之一: $2 - \frac{2}{n}$, $2\left(1 - \frac{1}{n}\right)$, $\frac{\lambda(n-1)}{n}$, $\lambda - \frac{\lambda}{n}$, $\lambda\left(1 - \frac{1}{n}\right)$

三、(15 分)

(1) 记 A_k 表示事件“第 k 次试验成功”。则

$$P(A_{k+1}|A_k) = \frac{2}{3}, \quad P(A_{k+1}|\overline{A_k}) = \frac{1}{6}$$

于是

$$\begin{aligned} P(A_{k+1}) &= P(A_k)P(A_{k+1}|A_k) + P(\overline{A_k})P(A_{k+1}|\overline{A_k}) = P(A_k) \times \frac{2}{3} + (1 - P(A_k)) \times \frac{1}{6} \\ &= \frac{1}{6} + P(A_k) \times \frac{1}{2} \end{aligned}$$

因此

$$P(A_2) = \frac{1}{6} + \frac{1}{3} \times \frac{1}{2} = \frac{1}{3}$$

利用上述递推关系及数学归纳法可以证明对任意正整数 k , $P(A_k) = \frac{1}{3}$

(2)

$$\begin{aligned} P(A_2|A_3) &= \frac{P(A_2 A_3)}{P(A_3)} \\ &= \frac{P(A_2)P(A_3|A_2)}{P(A_3)} \\ &= P(A_3|A_2) = \frac{2}{3} \end{aligned}$$

(3) $P(X=1) = P(A_1) = \frac{1}{3}$;

当 $k > 1$ 时,

$$\begin{aligned}
P(X=k) &= P(\overline{A_1} \overline{A_2} \cdots \overline{A_{k-1}} A_k) \\
&= P(\overline{A_1}) P(\overline{A_2} | \overline{A_1}) \cdots P(\overline{A_{k-1}} | \overline{A_1} \overline{A_2} \cdots \overline{A_{k-2}}) P(A_k | \overline{A_1} \overline{A_2} \cdots \overline{A_{k-1}}) \\
&= P(\overline{A_1}) P(\overline{A_2} | \overline{A_1}) \cdots P(\overline{A_{k-1}} | \overline{A_{k-2}}) P(A_k | \overline{A_{k-1}}) \\
&= \frac{2}{3} \times \left(\frac{5}{6}\right)^{k-2} \times \frac{1}{6} = \frac{1}{9} \times \left(\frac{5}{6}\right)^k
\end{aligned}$$

A 四(B 五)、(20 分)

(1)

$$\begin{aligned}
F_X(t) &= P(X \leq t) \\
&= 1 - P(X > t) \\
&= 1 - P(X_1 > t, X_2 > t) \\
&= 1 - P(X_1 > t) P(X_2 > t) \\
&= 1 - e^{-(\mu_1 + \mu_2)t}, \quad (t \geq 0)
\end{aligned}$$

即 X 服从指数分布 $\text{Exp}(\mu_1 + \mu_2)$ 。

(2)

解法 1:

$$\begin{aligned}
P(X_1 < X_2) &= \iint_{x < y} f_{X,Y}(x, y) dx dy \\
&= \int_0^{+\infty} \int_x^{+\infty} \mu_1 e^{-\mu_1 x} \mu_2 e^{-\mu_2 y} dy dx \\
&= \frac{\mu_1}{\mu_1 + \mu_2} \int_0^{+\infty} (\mu_1 + \mu_2) e^{-(\mu_1 + \mu_2)x} \int_x^{+\infty} \mu_2 e^{-\mu_2(y-x)} dy dx \\
&= \frac{\mu_1}{\mu_1 + \mu_2}
\end{aligned}$$

解法 2:

令 $\begin{cases} U = X_1 - X_2 \\ V = X_2 \end{cases}$ 。故

$$p_{U,V}(u, v) = \mu_1 e^{-\mu_1(u+v)} 1_{u+v>0} \cdot \mu_2 e^{-\mu_2 v} 1_{v>0}$$

进而,

$$p_U(u) = \int_{\max(0, -u)}^{+\infty} \mu_1 \mu_2 e^{-\mu_1 u} e^{-(\mu_1 + \mu_2)v} dv = \begin{cases} \frac{\mu_1 \mu_2}{\mu_1 + \mu_2} e^{-\mu_1 u}, & u > 0; \\ \frac{\mu_1 \mu_2}{\mu_1 + \mu_2} e^{\mu_2 u}, & u < 0. \end{cases}$$

因此

$$P(X_1 < X_2) = P(U < 0) = \int_{-\infty}^0 \frac{\mu_1 \mu_2}{\mu_1 + \mu_2} e^{\mu_2 u} du$$

$$= \frac{\mu_1}{\mu_1 + \mu_2}$$

解法 3:

令 $\begin{cases} W = X_1 / X_2 \\ V = X_2 \end{cases}$ 。故

$$p_{W,V}(w,v) = \mu_1 \mu_2 e^{-\mu_1 w v - \mu_2 v} 1_{w>0, v>0}$$

进而,

$$p_W(w) = \frac{\mu_1 \mu_2}{(\mu_1 w + \mu_2)^2} 1_{w>0}$$

因此

$$P(X_1 < X_2) = P(W < 1) = \int_0^1 \frac{\mu_1 \mu_2}{(\mu_1 w + \mu_2)^2} dw$$

$$= \frac{\mu_1}{\mu_1 + \mu_2}$$

(3)

解法 1:

$$F_{X_1|X_1 < X_2}(t) = P(X_1 \leq t | X_1 < X_2)$$

$$= 1 - P(X_1 > t | X_1 < X_2)$$

$$= 1 - \frac{P(t < X_1 < X_2)}{P(X_1 < X_2)}$$

$$= 1 - \frac{\mu_1 + \mu_2}{\mu_1} \int_{\max(t,0)}^{+\infty} \int_x^{+\infty} \mu_1 e^{-\mu_1 x} \mu_2 e^{-\mu_2 y} dy dx$$

$$= 1 - \int_{\max(t,0)}^{+\infty} (\mu_1 + \mu_2) e^{-(\mu_1 + \mu_2)x} \int_x^{+\infty} \mu_2 e^{-\mu_2(y-x)} dy dx$$

$$= 1 - e^{-(\mu_1 + \mu_2)\max(t,0)} = \begin{cases} 1 - e^{-(\mu_1 + \mu_2)t}, & t \geq 0; \\ 0, & t < 0. \end{cases}$$

即在已知 $X_1 < X_2$ 的条件下, X_1 服从指数分布 $Exp(\mu_1 + \mu_2)$, 从而

$$E(X_1 | X_1 < X_2) = \frac{1}{\mu_1 + \mu_2}$$

解法 2:

$$\begin{aligned}
 p_{X_1|X_1 < X_2}(t) &= \frac{P(X_1 \in dt, X_1 < X_2)}{P(X_1 < X_2)} \\
 &= \frac{\int_t^{+\infty} \mu_1 \mu_2 e^{-\mu_1 t - \mu_2 y} dy}{\frac{\mu_1}{\mu_1 + \mu_2}} \\
 &= (\mu_1 + \mu_2) e^{-(\mu_1 + \mu_2)t}
 \end{aligned}$$

所以,

$$\begin{aligned}
 E(X_1 | X_1 < X_2) &= \int_0^{+\infty} t p_{X_1|X_1 < X_2}(t) dt \\
 &= \frac{1}{\mu_1 + \mu_2}
 \end{aligned}$$

解法 3:

$$\begin{aligned}
 E(X_1 | X_1 < X_2) &= \frac{E(X_1 1_{X_1 < X_2})}{P(X_1 < X_2)} \\
 &= \frac{\int_0^{+\infty} \left(\int_t^{+\infty} t \cdot \mu_1 e^{-\mu_1 t} \cdot \mu_2 e^{-\mu_2 y} dy \right) dt}{\frac{\mu_1}{\mu_1 + \mu_2}} \\
 &= \frac{1}{\mu_1 + \mu_2}
 \end{aligned}$$

(4)

证法 1:

$$\begin{aligned}
 P(X_1 \leq t, X_1 < X_2) &= P(X_1 < X_2) P(X_1 \leq t | X_1 < X_2) \\
 &= \frac{\mu_1}{\mu_1 + \mu_2} (1 - e^{-(\mu_1 + \mu_2)t}) \\
 &= P(X_1 < X_2) P(X \leq t)
 \end{aligned}$$

证法 2:

$$\begin{aligned}
 p_{I,X}(1,t) &= p_{I,X_1}(1,t) \\
 &= P(X_1 < X_2 | X_1 = t) p_{X_1}(t) \\
 &= P(t < X_2 | X_1 = t) p_{X_1}(t) \\
 &= P(t < X_2) p_{X_1}(t) \\
 &= e^{-\mu_2 t} \cdot \mu_1 e^{-\mu_1 t} = \frac{\mu_1}{\mu_1 + \mu_2} \cdot (\mu_1 + \mu_2) e^{-(\mu_1 + \mu_2)t} \\
 &= P(I=1) p_X(t)
 \end{aligned}$$

A 五(B 四)、(15 分)

(1)

$$F_Z(t) = P(Z \leq t) = P(-\ln X \leq t) = P(X \geq e^{-t}) = \begin{cases} 1 - e^{-t}, & t \geq 0; \\ 0, & t < 0. \end{cases}$$

故 $Z \sim \text{Exp}(1)$

(2)

解法 1: 卷积公式

$$\begin{aligned} f_{X+Y}(t) &= \int_{-\infty}^{+\infty} f_X(x) f_Y(t-x) dx \\ &= \int_0^1 e^{-(t-x)} 1_{t-x>0} dx \\ &= 1_{t>0} e^{-t} \int_0^{\min(1,t)} e^x dx \\ &= 1_{t>0} e^{-t} (e^{\min(1,t)} - 1) \\ &= \begin{cases} e^{-t}(e-1), & t > 1; \\ 1 - e^{-t}, & 0 < t < 1; \\ 0, & t < 0. \end{cases} \end{aligned}$$

解法 2: 先求概率分布函数

$$\begin{aligned} F_{X+Y}(t) &= P(X+Y \leq t) \\ &= \begin{cases} \int_0^1 \int_0^{t-x} f_X(x) f_Y(y) dy dx, & t \geq 1; \\ \int_0^t \int_0^{t-x} f_X(x) f_Y(y) dy dx, & 0 \leq t < 1; \\ 0 & t < 0. \end{cases} \\ &= \begin{cases} \int_0^1 1 - e^{x-t} dx = 1 - e^{-t}(e-1) & t \geq 1; \\ \int_0^t 1 - e^{x-t} dx = t + e^{-t} - 1 & 0 \leq t < 1; \\ 0 & t < 0. \end{cases} \end{aligned}$$

再求导

得概率密度函数 (见左边)。

(3)

$$\begin{aligned}
P\left(Y \leq 1 \mid X \leq e^{\frac{-(Y-1)^2}{2}}\right) &= \frac{P\left(Y \leq 1, X \leq e^{\frac{-(Y-1)^2}{2}}\right)}{P\left(X \leq e^{\frac{-(Y-1)^2}{2}}\right)} \\
&= \frac{\iint_{\substack{0 < x < 1 \\ x \leq e^{\frac{-(y-1)^2}{2}}}} e^{-y} 1_{y>0} dx dy}{\iint_{\substack{0 < x < 1 \\ x \leq e^{\frac{-(y-1)^2}{2}}}} e^{-y} 1_{y>0} dx dy} = \frac{\int_0^1 e^{-y} \left(\int_0^{e^{\frac{-(y-1)^2}{2}}} dx \right) dy}{\int_0^{+\infty} e^{-y} \left(\int_0^{e^{\frac{-(y-1)^2}{2}}} dx \right) dy} \\
&= \frac{\int_0^1 e^{-y} e^{\frac{-(y-1)^2}{2}} dy}{\int_0^{+\infty} e^{-y} e^{\frac{-(y-1)^2}{2}} dy} = \frac{e^{-\frac{1}{2}} \sqrt{2\pi} \int_0^1 \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy}{e^{-\frac{1}{2}} \sqrt{2\pi} \int_0^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy} = \frac{\int_0^1 \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy}{\int_0^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy} \\
&= \frac{\Phi(1) - \Phi(0)}{1 - \Phi(0)} = 2\Phi(1) - 1.
\end{aligned}$$

六、(13分)

(1) 由

$$\begin{aligned}
EX &= \int_1^{+\infty} x \cdot \frac{1}{\lambda} e^{-\frac{(x-1)}{\lambda}} dx \\
&= 1 + \lambda
\end{aligned}$$

得

$$\lambda = EX - 1$$

用样本均值 $\bar{X}$ 代换上式中的总体均值 EX ，得到 λ 的矩估计量：

$$\hat{\lambda}_M = \bar{X} - 1$$

由

$$\begin{aligned}
E\hat{\lambda}_M &= E(\bar{X} - 1) \\
&= EX - 1 \\
&= \lambda, \quad \forall \lambda > 0
\end{aligned}$$

知 $\hat{\lambda}_M = \bar{X} - 1$ 是 λ 的无偏估计。

(2) 由

$$Var(X) = \lambda^2$$

知

$$E(S^2) = \text{Var}(X) = \lambda^2$$

由中心极限定理, 当 n 充分大时, $\bar{X}$ 近似服从如下正态分布

$$\bar{X} \sim N\left(\lambda + 1, \frac{\lambda^2}{n}\right)$$

七、(12分)

(1) 由 $\sqrt{\frac{9}{S^2}}(\bar{X} - \mu) \sim t(8)$, $P\left(\sqrt{\frac{9}{S^2}}(\bar{X} - \mu) \leq t_{0.95}(8)\right) = 0.95$, 得 μ 的 95%置信度的置信下界为

$$\bar{x} - t_{0.95}(8)\sqrt{\frac{s^2}{9}} = 7.68 - t_{0.95}(8) \times \sqrt{\frac{0.64}{9}} = 7.68 - \frac{0.8}{3} \times 1.860 = -7.184$$

(2) 因

$$P_{\mu \geq 8}\left(\sqrt{\frac{9}{S^2}}(\bar{X} - 8) \leq t_{0.05}(8)\right) \leq P_{\mu \geq 8}\left(\sqrt{\frac{9}{S^2}}(\bar{X} - \mu) \leq t_{0.05}(8)\right) = 0.05$$

故拒绝域为 $\sqrt{\frac{9}{S^2}}(\bar{X} - 8) \leq t_{0.05}(8) = -t_{0.95}(8) = -1.860$ 。代入样本值

$$\sqrt{\frac{9}{0.64}}(7.68 - 8) = -0.32 \times \frac{3}{0.8} = -1.2 > -1.860$$

这表明样本尚未落入拒绝域, 故不能拒绝原假设 $\mu \geq 8$ 。